

Modeling and Predicting Artist Career Decline in Music Streaming

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Abstract—We present a two-stage data mining framework for predicting artist popularity decline using longitudinal Spotify data. From 32,828 raw tracks, we built artist-year profiles for 1,094 artists (3,474 labeled observations). Stage one applies Ward linkage clustering and discovers eight career archetypes with decline rates ranging from 16% to 53% ($\chi^2 = 109.94$, $p < 0.0001$, Cramér’s $V = 0.178$). Stage two trains supervised classifiers to predict a 10-point popularity drop in the following year. Gradient Boosting achieves 92.0% recall on the held-out test set (95% CI: [87.5%, 96.2%]), catching 127 of 138 real decline cases. Current popularity and recent yearly change are the two dominant predictors. A data ceiling limits further improvement. Advancing beyond this ceiling requires off-platform signals that Spotify’s dynamic popularity score cannot encode.

Index Terms—music streaming, artist popularity decline, career lifecycle, agglomerative clustering, gradient boosting, Spotify

I. INTRODUCTION

Spotify’s 0–100 popularity score reflects recent streaming activity across 600M+ users, providing a continuous measure of artist career momentum [6]. Despite this richness, predicting when an artist’s popularity will start to decline has received little attention in the data mining literature.

This matters in practice. Talent managers and A&R teams [7] need early warning before decline begins because once a career has fallen, the window for intervention (contract renegotiation, promotional campaigns, artistic strategy) may have closed.

We address two questions: (1) can we identify meaningful career archetypes from longitudinal Spotify data, and (2) can we predict next-year decline from career trajectory features? We make four contributions:

- A longitudinal profiling pipeline producing 11 time-aware features
- Eight statistically validated career archetypes via agglomerative clustering
- A supervised decline prediction framework with expanding-window CV and F_2 -optimal threshold tuning
- An archetype-level error analysis linking model failures to career structure

II. RELATED WORK

Arakelyan et al. [1] predicted artist career milestones from concert network data but did not model year-over-year stream-

ing changes. Gulmatico et al. [2] predicted single-track Spotify popularity from audio features but did not study longitudinal artist trajectories. Lee and Lee [4] defined eight popularity metrics from long-term Billboard chart data and showed that audio features predict popularity better than chance, but acknowledged a persistent ceiling in predictive accuracy, which is a finding consistent with our results. Webster and Zander [3] showed that critic and fan ratings decline over time using subjective ratings, not streaming prediction. To our knowledge, this is the first work combining unsupervised lifecycle clustering with supervised year-over-year decline prediction on Spotify data.

III. DATA PREPARATION AND FEATURE ENGINEERING

A. Dataset

We use the 30,000 Spotify Songs dataset from Kaggle [5] (32,828 tracks). We filtered tracks from **1990–2020** since that is where most of the data lied. After deduplication and filtering for artists with ≥ 3 active years, the working set is **1,094 artists**, collapsed to 4,568 artist-year profiles. The dataset is playlist-based and skews toward curated, popular music.

B. Design Decisions

Three threshold choices shape the analysis (Table I).

TABLE I
KEY DESIGN DECISIONS

Decision	Value	Rationale
Min. career length	3 yr	2 yr gives only one change value
Decline threshold	10 pts	5 pts is noise; 15 pts severe imbalance
Train/test cutoff	2016	Years ≤ 2016 contain $\sim 80\%$ of observations

C. Feature Engineering

Eleven time-aware features are built using expanding-window operations (Table II). Every feature for year t uses only data from years $\leq t$. *yearly_change* is divided by the actual year gap between releases, preventing multi-year gaps from being misrepresented as single-year drops.

TABLE II
ENGINEERED CAREER TRAJECTORY FEATURES

Feature	What it measures
<i>yearly_change</i>	How much popularity rose or fell vs. last year
<i>variability</i>	How much popularity bounces around year to year
<i>career_age</i>	How many years the artist has been active
<i>peak_popularity</i>	Highest popularity the artist has ever reached
<i>gap_from_peak</i>	How far below their all-time high they currently are
<i>years_since_peak</i>	How long ago they hit their highest point
<i>career_slope</i>	Average direction of popularity over their whole career
<i>release_consistency</i>	Did they release music last year? (1 = yes, 0 = no)
<i>track_count_change</i>	Are they releasing more or fewer songs than last year?
<i>momentum</i>	Are they doing better or worse than their own average?
<i>recent_ratio</i>	Current popularity as a fraction of their personal best

D. Decline Label and Split

A decline is a drop of ≥ 10 points in the following year. This produces 3,474 labeled rows (25.6% decline rate). Training: years ≤ 2016 (2,639 rows, 28.5% decline). Test: years > 2016 (835 rows, 16.5% decline). The lower test rate reflects Spotify’s recency weighting since older tracks lose streaming share over time, making historical drops appear larger.

IV. CAREER LIFECYCLE CLUSTERING

A. Method

One summary row per artist is built by aggregating each artist’s full career history into eight features: career length (number of active years), mean popularity, peak popularity (career maximum), minimum popularity (career minimum), mean yearly change, mean variability, mean momentum, and mean recent ratio. Missing values are filled with the column median before scaling. All eight features are then Z-score standardized to mean 0 and standard deviation 1. Decline rate is computed per artist but excluded from clustering because it is used only afterward to validate and interpret the resulting archetypes.

Ward linkage agglomerative clustering is selected because it penalizes merges that increase within-cluster variance, making it well-suited for trajectory data where careers exist on a continuous spectrum. Three alternatives were tested and discarded: single/average linkage degenerated to 1093 vs. 1 splits; complete linkage isolated the same 12-artist outlier at every k ; DBSCAN labeled 573 of 1,094 artists as noise (silhouette = -0.136).

Silhouette scores over $k = 2-8$ (Table III) peak at $k = 8$ (score = 0.219). This is consistent with the fact that careers can exist on a spectrum rather than in discrete groups.

TABLE III
SILHOUETTE SCORES BY NUMBER OF CLUSTERS

k	Silhouette
2	0.149
3	0.179
4	0.214
5	0.215
6	0.194
7	0.209
8	0.219

B. The Eight Archetypes

Table IV summarizes all eight archetypes by size and decline rate. Table V gives the three most diagnostic feature values for each. Fig. 1 shows the full normalized feature profile per cluster. Figs. 2 and 3 show one representative artist per archetype.

TABLE IV
CAREER LIFECYCLE ARCHETYPES — SIZE AND DECLINE RATE

Archetype	n	Decline Rate
Legacy Faders	49	53%
Momentum Lost	116	40%
Low-Mid Steady	264	24%
Mainstream Stable	341	24%
Long-Running Stable	105	24%
Low Current Visibility	70	19%
Established High-Performers	54	17%
Catalog Risers	95	16%

TABLE V
KEY FEATURE VALUES PER ARCHETYPE

Archetype	Pop.	Momentum	Yly Δ
Mainstream Stable	58.5	+0.12	+1.6
Long-Running Stable	46.9	+1.40	+1.9
Established High-Performers	41.9	+6.78	+12.6
Low-Mid Steady	40.2	-0.14	-1.0
Momentum Lost	40.2	-2.97	-2.4
Legacy Faders	30.8	0.00	-25.4
Catalog Risers	29.6	+0.47	+23.6
Low Current Visibility	16.1	-0.60	-2.1

Three archetypes stand out. *Momentum Lost* ($n = 116$, 40% decline rate) is the most actionable group for talent managers. Artists such as Ariana Grande and The Weeknd reached real peaks but now show the most negative momentum of any cluster (-2.97), meaning they are consistently underperforming their own career averages. Early intervention here has the clearest justification. *Low Current Visibility* ($n = 70$) is the most misunderstood group. Artists such as Bruce Springsteen,



Fig. 1. Normalized feature means per cluster (0 = lowest, 1 = highest across all clusters). Established High-Performers lead on momentum and peak popularity. Legacy Faders lead on variability and decline rate. Low Current Visibility shows the lowest popularity scores of all groups.

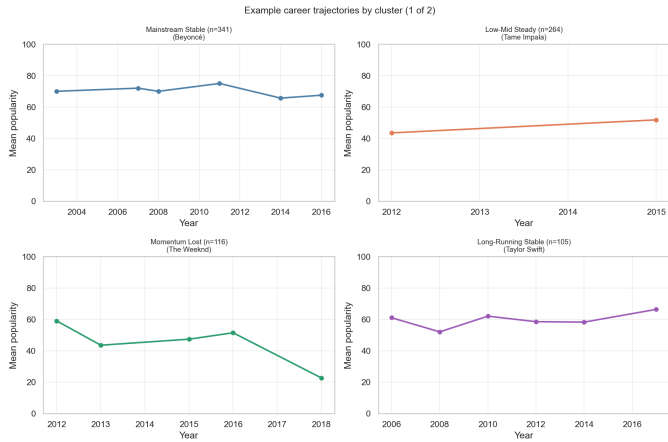


Fig. 2. Representative trajectories (archetypes 1–4). Beyoncé (Mainstream Stable) holds steady near 70 for over a decade. Tame Impala (Low-Mid Steady) drifts slowly upward. The Weeknd (Momentum Lost) peaks but then falls sharply. Taylor Swift (Long-Running Stable) sustains mid-high popularity across the her visible career window.

Marvin Gaye, and Erykah Badu appear here not because their careers were weak, but because Spotify’s popularity score down-weights older catalogs as newer music accumulates streams. This cluster illustrates the core data limitation of this study. *Legacy Faders* ($n = 49$, 53% decline rate) has the highest risk of any cluster. These are classic acts like Metallica and The Rolling Stones whose entire visible Spotify window is a later career snapshot. The model sees them declining because that is all the data shows.

The remaining five archetypes. *Mainstream Stable* ($n = 341$) is the largest group and the closest to a baseline — popular, consistent, and neither meaningfully rising nor falling. Artists like Beyoncé, Billie Eilish, and Bruno Mars anchor this cluster. *Low-Mid Steady* ($n = 264$) sits in the middle of the popularity range with a slight downward drift but no dramatic trajectory shift. Artists such as Imagine Dragons, Michael Jackson, and Tame Impala are slowly fading without

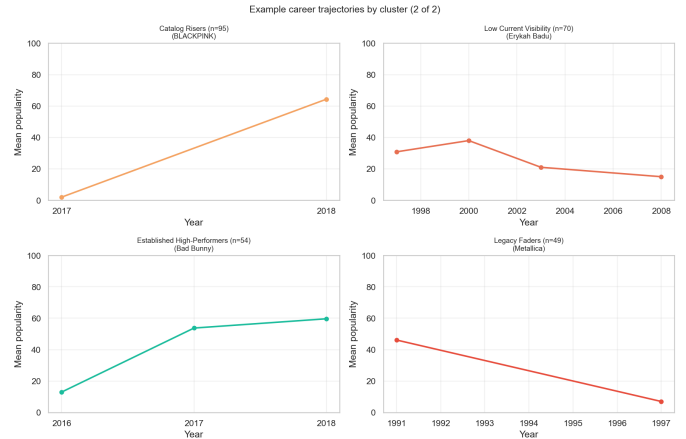


Fig. 3. Representative trajectories (archetypes 5–8). BLACKPINK (Catalog Risers) surges from near zero to 65 in two years and holds; only two data points are visible because they entered the dataset late and their most recent year is dropped when constructing the decline label (this is expected, not a data error). Erykah Badu (Low Current Visibility) scores low not from career weakness but from Spotify’s recency weighting on older catalogs. Bad Bunny (Established High-Performers) climbs steeply and is still rising. Metallica (Legacy Faders) drops sharply across only the earliest Spotify window visible for that act.

a clear turning point. *Long-Running Stable* ($n = 105$) has the longest careers of any cluster and is still performing above its own historical average. This includes artists such as Drake, Eminem, and Shakira who have sustained relevance over many years. *Catalog Risers* ($n = 95$) are the fastest growing group, currently at or near their personal peak with the lowest decline rate of any cluster (16%). This includes acts such as BLACKPINK, Jorja Smith, and Cigarettes After Sex whose Spotify presence is still climbing. *Established High-Performers* ($n = 54$) have the highest peak popularity and momentum of any cluster. These are artists such as Bad Bunny, BTS, and Post Malone who are still building toward their ceiling and show no signs of decline.

C. Statistical Validation

Chi-square test on artist-year-level decline labels confirms the association between archetype and decline probability: $\chi^2 = 109.94$, $p < 0.0001$, $df = 7$, Cramér’s $V = 0.178$. The weak V is consistent with the silhouette score of 0.219 because archetypes are real tendencies, not hard categories.

Fig. 4 shows that `mean_popularity`, `peak_popularity`, and `max_popularity` are correlated above 0.82, meaning the clustering is partially driven by a single underlying popularity dimension.

V. DECLINE PREDICTION

A. Setup

Metrics. PR-AUC evaluates ranking ability on the minority class without threshold dependence (random baseline = 0.165). F2 score is the deployment metric, weighting recall twice as heavily as precision. Missing a real decline costs more than a false alarm: promotional campaigns cost \$50K–\$1M+

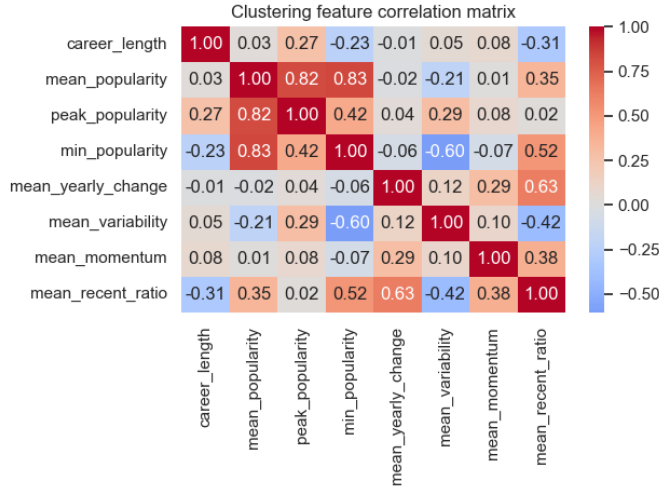


Fig. 4. Pearson correlation matrix for the eight clustering features. Three popularity-related features (mean, peak, max) are correlated above 0.82, reducing the effective dimensionality of the clustering.

and must launch before decline accelerates, while standard contracts run 6–9 years [7].

Baseline. A momentum rule (predict decline if yearly change < -5) achieves PR-AUC = 0.157, below the random baseline. Momentum alone has no predictive value.

Validation. Four expanding-window CV folds (test years 2013, 2014, 2015, 2016) are used for hyperparameter tuning and model selection. The inner split for threshold tuning holds out the most recent year in each fold’s training data, preserving temporal ordering.

B. Models

We train four models: Decision Tree (interpretability baseline), Random Forest, Gradient Boosting (HistGradientBoostingClassifier), and XGBoost. Best hyperparameters from grid search are shown in Table VI. All models address class imbalance through balanced weighting or equivalent scale adjustment. Missing first-year values are median-imputed on the training set only.

TABLE VI
BEST HYPERPARAMETERS FROM GRID SEARCH

Model	Key Parameters
Decision Tree	max_depth=4, balanced
Random Forest	200 trees, max_depth=None, min_leaf=10, balanced
Gradient Boosting	300 iter, lr=0.01, max_depth=3, min_leaf=20, balanced
XGBoost	300 trees, lr=0.01, max_depth=3, spw=2.51

C. Threshold Tuning

Each model’s threshold is tuned by maximizing F2 on the 2016 validation fold only, then fixed before the test set is opened. The F2-optimal threshold for Gradient Boosting is

0.369 (val F2 = 0.672). Fig. 5 shows the precision-recall-F2 tradeoff across thresholds (the chosen value favors recall over precision as required by the cost structure of the application).

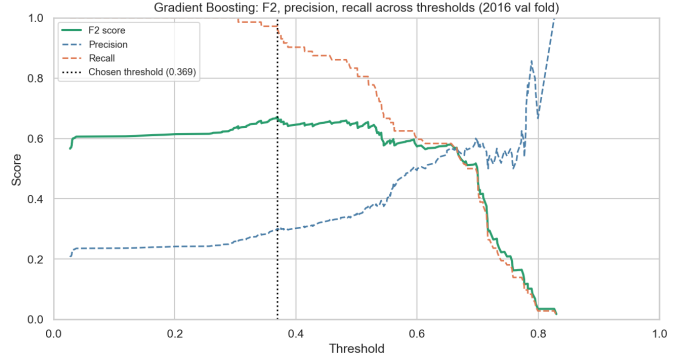


Fig. 5. Gradient Boosting precision, recall, and F2 across decision thresholds on the 2016 validation fold. The selected threshold of 0.369 (dashed line) maximizes F2, weighting recall twice as heavily as precision.

D. Results and Model Selection

Fig. 6 shows precision-recall curves for all models. Table VII shows full results.

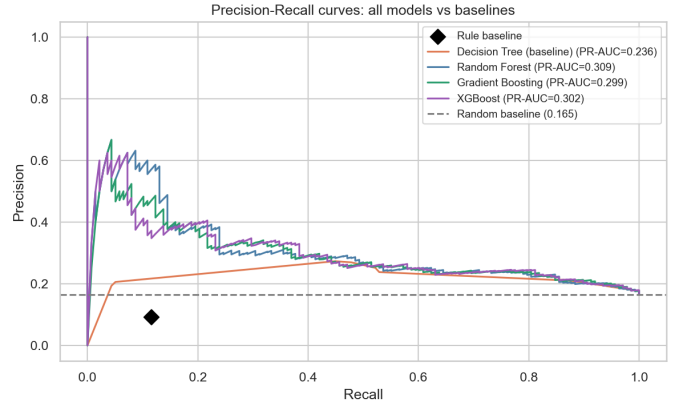


Fig. 6. Precision-recall curves. All models exceed the rule baseline (PR-AUC 0.157) and random baseline (0.165). The rule baseline point falls below the random baseline curve.

TABLE VII
MODEL RESULTS — CV AND HOLDOUT AT TUNED THRESHOLD

Model	PR-AUC	CV Rec.	CV F2	Test Rec.	Test F2
GB [†]	0.299	0.888	0.665	0.920	0.539
XGBoost	0.302	0.900	0.667	0.906	0.535
Rand. Forest	0.309	0.823	0.660	0.855	0.532
Dec. Tree	0.236	—	—	0.848	0.530
Rule base	0.157	—	—	—	—
Rand. base	0.165	—	—	—	—

[†]Selected model (Gradient Boosting).

Model selection. During hyperparameter tuning, the unified CV F2 criterion (cv_score_f2) ranked Random Forest first

(0.677 vs. 0.670 for Gradient Boosting and 0.668 for XGBoost), making it the initial pre-committed selection. Note that the CV recall and CV F2 columns in Table VII reflect per-fold re-evaluation at each model’s tuned threshold (a different measurement than the tuning-grid F2 used for selection). The margin across all three ensemble models was 0.007, so effectively a tie. XGBoost leads on both CV recall (0.900) and CV F2 (0.667) at the tuned threshold, but Gradient Boosting achieves the highest test recall (0.920 vs. 0.906 and 0.855), suggesting it generalizes more reliably to unseen data. Because the project objective is maximizing recall of real declines, Gradient Boosting is selected. This deviation from the pre-committed criterion is justified by the effective tie in CV performance and is documented explicitly here.

Final model. At threshold 0.369, Gradient Boosting achieves TP = 127, FN = 11, FP = 499, TN = 198. Fig. 7 shows confusion matrices for the final model and Random Forest as reference. Bootstrap confidence intervals ($n_{boot} = 1000$): PR-AUC 0.299 [0.239, 0.381], recall 0.920 [0.875, 0.962], precision 0.203 [0.172, 0.236], F2 0.539 [0.490, 0.586]. The cost is 3.9 false alarms per correct detection, so roughly 1 in 5 flagged artists actually declines.

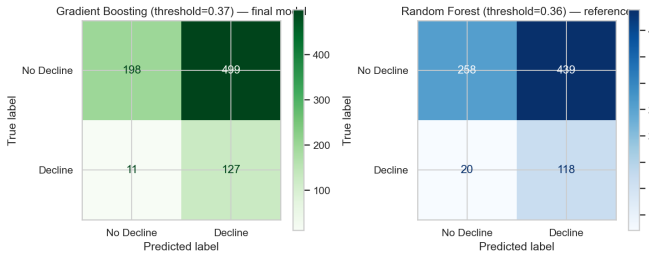


Fig. 7. Confusion matrices for Gradient Boosting (final, threshold = 0.369, left) and Random Forest (reference, threshold = 0.364, right). High recall is achieved at the cost of many false positives, consistent with the F2 objective.

E. Feature Importance

HistGradientBoostingClassifier does not expose native feature importances, so permutation importance is used throughout. Fig. 8 shows permutation importance on test and train sets. Mean popularity dominates (test: 0.112, train: 0.222). Yearly change is the only other feature with meaningful test importance (0.045). All other features contribute negligible independent signal.

Ablation. Retraining on only *mean_popularity* and *yearly_change* drops PR-AUC from 0.299 to 0.281. Adding *peak_popularity* as a third feature produces no further gain (PR-AUC still 0.281), empirically confirming the high correlation between mean and peak popularity.

F. Error Analysis by Archetype

Fig. 9 shows the false negative rate per cluster. Low Current Visibility has the highest miss rate (5.6%) and zero false positives, so the model almost never flags any artist in this group, whether they are actually declining or not. The reason is that *mean_popularity* is the model’s strongest signal, and

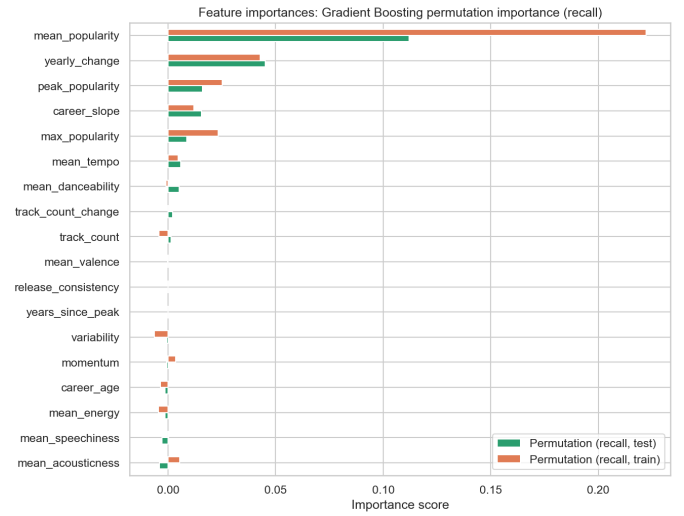


Fig. 8. Permutation importance (recall) on test and train sets. Mean popularity and yearly change carry nearly all predictive signal. The train/test gap on mean popularity is expected and does not indicate overfitting.

these artists score so low (mean = 16.1) that they never cross the model’s decision boundary. For example, an artist dropping from 75 to 63 popularity looks alarming to the model, but an artist dropping from 15 to 3 looks the same as someone who was always inactive. The same 12-point drop has a completely different response from the model.

Established High-Performers (78.9% false positive rate) and Mainstream Stable (72.0%) account for the majority of the 499 false positives. BTS in 2018 (prob = 0.815, no decline) and J Balvin in 2019 (prob = 0.803, no decline) are indistinguishable from true positives such as Zedd in 2018 (prob = 0.849, declined) and Mark Ronson in 2018 (prob = 0.843, declined) using Spotify data alone.

Per-archetype ablation shows that the full 18-feature model gains most over the 2-feature baseline (*mean_popularity* + *yearly_change*) for Momentum Lost (+16.7 recall pts), Mainstream Stable (+8.7 pts), and Long-Running Stable (+8.3 pts). Two features are sufficient for Catalog Risers and Established High-Performers.

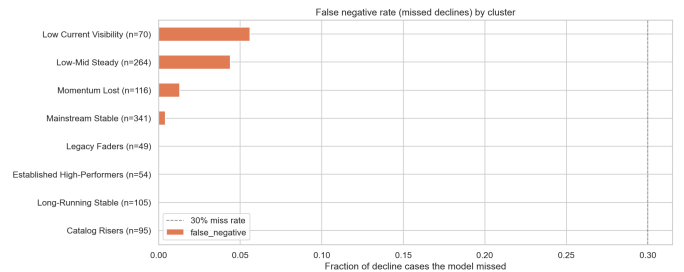


Fig. 9. False negative rate by career archetype. Low Current Visibility has the highest miss rate (5.6%) because low-popularity artists fall far below the model’s decision boundary regardless of trajectory. Momentum Lost benefits most from the full feature set (+16.7 recall points above the 2-feature baseline).

VI. DISCUSSION

Spotify popularity scores impose a hard ceiling on decline prediction. The model identifies trajectory direction reliably but cannot determine whether a trajectory will reverse. Off-platform events (label changes, controversies, comebacks, tours) are invisible to any feature derivable from streaming data. The dominant signal, current popularity and recent yearly change, reflects this ceiling directly. Sixteen of eighteen engineered features add only marginal collective value beyond those two, because Spotify scores are snapshots of recent activity and cannot encode the events that most directly drive career change. At 3.9 false alarms per correct detection, the model is best suited for narrowing a watchlist, not replacing human judgment.

VII. CONCLUSION

We present a two-stage framework for modeling artist career decline in music streaming. Ward linkage clustering discovers eight statistically distinct archetypes with decline rates ranging from 16% to 53%. Gradient Boosting with F2-optimal threshold tuning catches 92.0% of real declines (95% CI: [87.5%, 96.2%]). The most actionable finding is that Momentum Lost artists, those with the most negative momentum, are identifiable before decline accelerates, giving talent managers a concrete early warning signal. Three directions could raise the current ceiling: (1) incorporating social media, press coverage, and tour data; (2) training separate per-archetype models; and (3) using monthly or quarterly profiles to expose within-year shifts that annual averaging discards. Without off-platform signals, the data ceiling identified here will persist regardless of model choice.

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